

# Labour Market Forecasting (ROA)

## MSc thesis defence

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- Project Onderwijs en Arbeidsmarkt (POA) from ROA forecasts labour market needs with 6 years horizon.
- Problem with the current forecasting model.
  - Model used planned for explaining cointegration, not forecasting. No SR dynamics.
- Shift in the thesis to modify and apply other forecasting techniques to POA project.

- Data provided by CBS, from the Labour Force Survey (LFS).
- Years span from 1996 to 2022 and around 8.5 million people in the dataset.
- 21 industrial sectors and 39 occupational segments → sector-segment combinations.
- Changes in the classification by CBS over the years → structural breaks.
- COVID and war in Ukraine had an impact on the labour market → larger errors in later years.

Two different approaches for forecasting:

- ① Cörvers and Dupuy (2010) (SDOLS):
  - No SR dynamics and no ECM.
  - Panel structure → allows sectoral effects.
  - Originally only for explaining not forecasting.
- ② Stock and Watson (1993):
  - ECM and SR dynamics.
  - Single equations → no interaction between sectors.
  - Variation (SW-Coint).

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# SDOLS forecast equations

- ① Employment equation from Cörvers and Dupuy (2010)<sup>1</sup>:

$$\ln L_{ijt} = a_{ij} + b_t + \underline{x}'_{it} (\underline{d}_i + \underline{d}_j) + e_{ijt}. \quad (1)$$

- ② Log employment  $\ln L_{ijt}$  is residualized.<sup>2</sup>

$$\widetilde{\ln L_{ijt}} = \ln L_{ijt} - \widehat{\ln L_{ijt}}. \quad (2)$$

- ③ Explanatory variables are residualized.  
④ Residualized log employment regressed on residualized explanatory variables.  
⑤ Predicted value of  $\widehat{\ln L_{ijt}}$ : Fitted value first-step employment plus fitted second-step residual log employment:

$$\widehat{\ln L_{ijt}} = \widehat{\ln L_{ijt}}^{(1)} + \widehat{\widetilde{\ln L_{ijt}}}^{(2)}. \quad (3)$$

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<sup>1</sup>  $\underline{x}'_{it}$  = explanatory variables in VA and in K

<sup>2</sup>  $i$  = sector,  $j$  = segment,  $t$  = time period/ year



- Extension of the SDOLS model with an ECM by Engle and Granger (1987).
- First the SDOLS model is performed, the ECM afterwards.
- The ECM allows to capture SR dynamics.
- Allows to keep panel structure and interaction between sectors.
- Variation with ECM applied only to cointegrated time series ("ECM-Hybrid-Coint").

# ECM-Hybrid equations

- 1 The SDOLS model is performed.
- 2 Error Correction term  $\widehat{EC}_{ijt}$  is calculated:

$$\widehat{EC}_{ijt} = \ln L_{ijt} - \widehat{\ln L_{ijt}}^{SDOLS}.$$

- 3 The Error Correction Model (ECM) is performed:

$$\begin{aligned} \Delta \ln L_{ijt} = & \alpha_{ij} + \rho_{ij} \Delta \ln L_{ij,t-1} \\ & + \gamma_{ij} \widehat{EC}_{ij,t-1} + \beta_{VA,ij} \Delta \ln VA_{ijt} + \beta_{K,ij} \Delta \ln K_{ijt} \\ & + u_{ijt}. \end{aligned} \tag{4}$$

In **blue** the only term that is already estimated in the ECM compared to the forecasting procedure.  
In **red** the only term that is not the forecasting procedure compared to the ECM equation.

- 1 The forecasted log employment  $\widehat{\ln L_{ijt}}$  value is calculated for period  $t$ :

$$\widehat{\ln L_{ijt}} = \widehat{\ln L_{ij,t-1}} + \widehat{\Delta \ln L_{ijt}}. \quad (5)$$

- 2 EC term is updated using the forecasted log employment  $\widehat{\ln L_{ijt}}$ , then the following year is forecasted.

The model:

$$\ln L_{ijt} = \alpha + \beta_{VA} \ln VA_{ijt} + \beta_K \ln K_{ijt} + \sum_{b=-1}^1 d_b^{VA} \Delta \ln VA_{ij,t-b} + \sum_{b=-1}^1 d_b^K \Delta \ln K_{ij,t-b} + \varepsilon_{ijt}. \quad (6)$$

Note:  $b \in \{-1, 0, 1\}$ .

$d_b^{VA}$  and  $d_b^K$ : Dynamic correction errors, take out endogeneity and cointegration.

## Added Error Correction Model:

$$\Delta \ln L_{ijt} = \mu + \gamma \hat{\varepsilon}_{ij,t-1} + \rho \Delta \ln L_{ij,t-1} + \delta_k \Delta \ln K_{ijt} + \delta_{va} \Delta \ln VA_{ijt} + u_{ijt}.$$

## Forecasting procedure:

$$\widehat{\ln L_{ijt}} = \ln \widehat{L_{ij,t-1}} + \widehat{\Delta \ln L_{ijt}}.$$

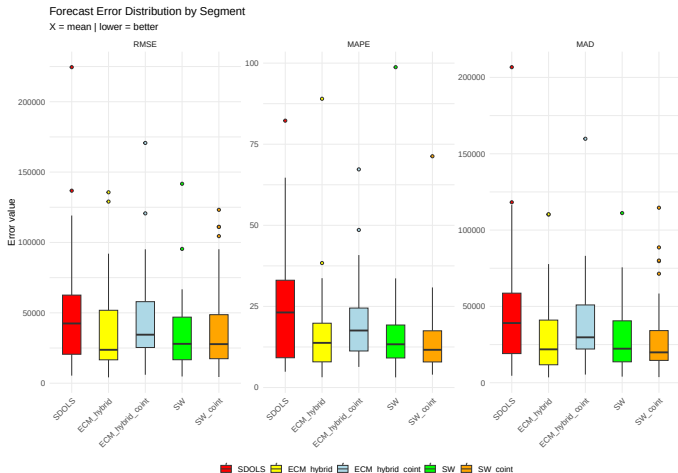
# Evaluation metrics

- Five different evaluation metrics used to compare performance of the models.
- RMSE, MAPE and MAD as regular metrics.
- rRMSE and rMAD as relative metrics with the SDOLS model as benchmark.
- Metrics for data aggregated on segments and on forecast years.
- Visualization as box plots for the segment data, points for the years.

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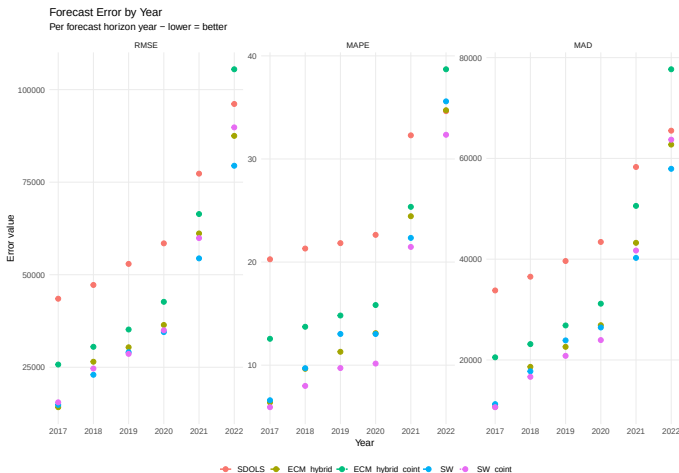
# Results regular metrics segments



- ECM-Hybrid and SW outperform SDOLS model.
- ECM-Hybrid-Coint performs worse than regular counterpart.



# Results regular metrics year



- SDOLS for near forecasts worst for first years, only better than ECM-Hybrid-CoInt for last years.
- Noticeable jump from 2020 to 2021.

# Relative metrics results segments

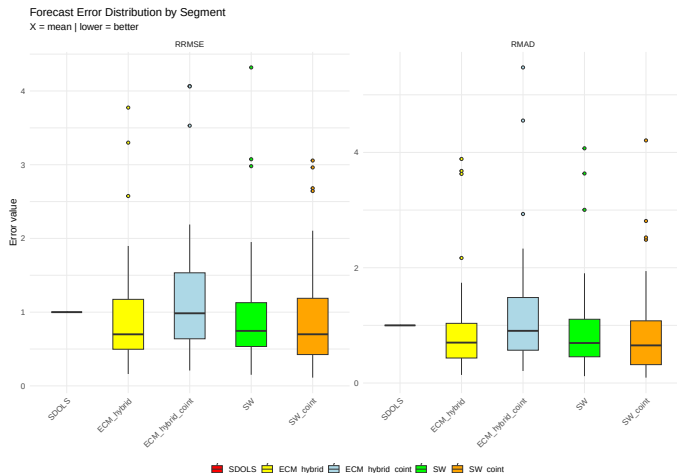


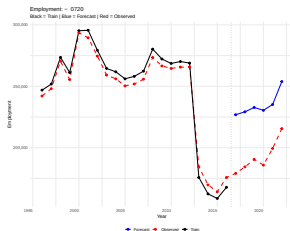
Figure: SDOLS as benchmark. Results show ratios.

- SW, SW-Coint and ECM-Hybrid perform better than benchmark.

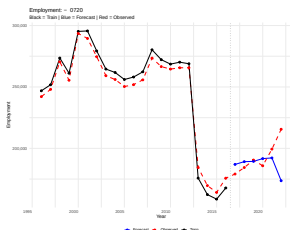
# Plots for segment 720

Forecasts for segment 720 (Science and engineering associate professionals) using three different models.

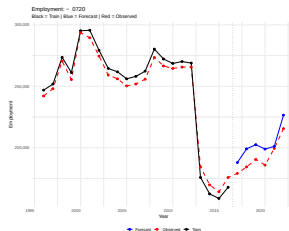
## SDOLS



## ECM-Hybrid



## SW



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# Key findings of the thesis

- ① The SDOLS model is outperformed by other models and promising results by other models.
- ② Noticeable differences among various segments.
  - From almost 50% significant in pairwise comparison for models to 10% significant.
- ③ Shift in results of pre- to in-COVID times.
- ④ Applying the ECM only on cointegrated time series does not guarantee improvements.

# Possible further research

- Further extension of the work on bootstrap CI.
- Further development of the whole forecast model to more detailed classification of data (occupational groups).
- Application of models to quarterly or monthly data, as few observations with annual data.
- Fine tuning of ECM-Hybrid-Coint model code → application of ML techniques to forecasting methods.

**Thank you very much for your attention and the great collaboration!**

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# Sources I



Cörvers, Frank and Arnaud Dupuy (Mar. 2010). “Estimating employment dynamics across occupations and sectors of industry”. en. In: *Journal of Macroeconomics* 32.1, pp. 17–27. ISSN: 01640704. DOI: 10.1016/j.jmacro.2009.09.007. URL: <https://linkinghub.elsevier.com/retrieve/pii/S016407040900069X> (visited on 12/12/2025).



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# SDOLS forecast equation 1

Model:

$$\ln L_{ijt} = a_{ij} + b_t + \underline{x}_{it}' (\underline{d}_i + \underline{d}_j) + e_{ijt} \quad (7)$$

Residualized log employment:

$$\widetilde{\ln L_{ijt}} = \ln L_{ijt} - \widehat{\ln L_{ijt}} \quad (8)$$

Residualized log employment regressed on residualized explanatory variables:

$$\ln L_{ijt} - \widehat{\ln L_{ijt}} = g_{ij} + (\underline{x}_{it} - \hat{\underline{x}}_{it})' (\underline{h}_i + \underline{h}_j) + \nu_{ijt} \quad (9)$$

# SDOLS forecast equation 2

Model as a whole:

$$\begin{aligned}\ln L_{ijt} = & \underbrace{a_{ij} + g_{ij} + \underline{d}'_i \underline{h}_j + \underline{d}'_j \underline{h}_i + b_t + \underline{x}'_{it}(\underline{h}_i + \underline{h}_j)}_{\text{Long-run relationship}} \\ & + \underbrace{\Delta \underline{x}'_{it} (\underline{c}_i + \underline{c}_j + \underline{e}_i''(\underline{h}_i + \underline{h}_j))}_{\text{Intra-sectorial dynamics}} \\ & + \underbrace{\left( \sum_{s \neq i} \Delta \underline{x}'_{st} \underline{e}_i^s \right)' (\underline{h}_i + \underline{h}_j)}_{\text{Inter-sectorial dynamics}} + \nu_{ijt}\end{aligned}\quad (10)$$

Forecasting procedure:

$$\widehat{\ln L_{ijt}} = \widehat{\ln L_{ijt}}^{(1)} + \widehat{\ln L_{ijt}}^{(2)} \quad (11)$$

# ECM Hybrid Equations

$$\widehat{\ln L_{ijt}}^{SDOLS} = \widehat{\ln L_{ijt}}^{(1)} + \widehat{\ln L_{ijt}}^{(2)} \quad (12)$$

Error Correction term  $\widehat{EC}_{ijt}$  is calculated:

$$\widehat{EC}_{ijt} = \ln L_{ijt} - \widehat{\ln L_{ijt}}^{SDOLS}.$$

The Error Correction Model (ECM) is performed:

$$\begin{aligned} \Delta \ln L_{ijt} = & \alpha_{ij} + \rho_{ij} \Delta \ln L_{ij,t-1} \\ & + \gamma_{ij} \widehat{EC}_{ij,t-1} + \beta_{VA,ij} \Delta \ln VA_{ijt} + \beta_{K,ij} \Delta \ln K_{ijt} \\ & + u_{ijt}. \end{aligned} \quad (13)$$

Forecasting procedure:

$$\widehat{\Delta \ln L_{ijt}} = \hat{\alpha}_{ij} + \hat{\rho}_{ij} \widehat{\Delta \ln L_{ij,t-1}} + \hat{\gamma}_{ij} \widehat{EC}_{ij,t-1} + \hat{\beta}_{VA,ij} \Delta \ln VA_{ijt} + \hat{\beta}_{K,ij} \Delta \ln K_{ijt}. \quad (14)$$

Updated EC term

$$\widehat{EC}_{ijt} = \widehat{\ln L_{ijt}} - \widehat{\ln L_{ijt}}^{SDOLS}. \quad (15)$$

The model:

$$\ln L_{ijt} = \alpha + \beta_{VA} \ln VA_{ijt} + \beta_K \ln K_{ijt} + \sum_{b=-1}^1 d_b^{VA} \Delta \ln VA_{ij,t-b} + \sum_{b=-1}^1 d_b^K \Delta \ln K_{ij,t-b} + \varepsilon_{ijt}. \quad (16)$$

Note:  $b \in \{-1, 0, 1\}$ .

$d_b^{VA}$  and  $d_b^K$ : Dynamic correction errors, take out endogeneity and cointegration.



## Added Error Correction Model:

$$\Delta \ln L_{ijt} = \mu + \gamma \hat{\varepsilon}_{ij,t-1} + \rho \Delta \ln L_{ij,t-1} + \delta_k \Delta \ln K_{ijt} + \delta_{va} \Delta \ln VA_{ijt} + u_{ijt}.$$

## Forecasting procedure:

$$\widehat{\ln L_{ijt}} = \widehat{\ln L_{ij,t-1}} + \widehat{\Delta \ln L_{ijt}}.$$

$$\hat{\varepsilon}_{ijt} = \widehat{\ln L_{ijt}} - \widehat{\ln L_{ijt}}^{LR}.$$

# Bootstrap plot

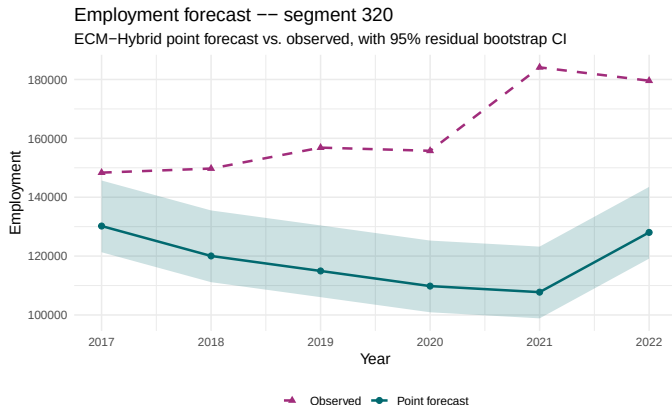


Figure: Bootstrap plot for segment 320 with the observed data to compare.

# Back transformation to levels

**Jensen correction and log-transformation** Correction factor ( $\alpha_0$ ):

$$\alpha_0 = \exp \left( \bar{\hat{u}} + \frac{1}{2} \widehat{\text{Var}}(\hat{u}) \right). \quad (17)$$

Employment on levels ( $\hat{L}_{ijt}$ ):

$$\hat{L}_{ijt} = \alpha_0 \exp \left( \widehat{\ln L}_{ijt} \right). \quad (18)$$

Rescaled employment to sectors:

$$\hat{L}_{ijt}^{adj} = \hat{L}_{ijt} \left( \frac{\text{empl}_{it}^{sec}}{\sum_j \hat{L}_{ijt}} \right). \quad (19)$$

**Aggregation to segments**

$$\hat{L}_{jt} = \sum_i \hat{L}_{ijt}^{adj}. \quad (20)$$